

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			$520 (1) \times 31.6 (1) = 16\,432 \text{ [km]}$ $16\,432 \times 8.9 \times 10^{-2} = 1\,462.4 \text{ [kg]} (1) \text{ answer}$ Alternatives $10 \times 31.6 = 316.0 (1) \text{ i.e. fuel in a week}$ $316.0 \times 8.9 \times 10^{-2} = 28.124 (1) \text{ i.e. CO}_2 \text{ in a week}$ $28.124 \times 52 = 1462.4 \text{ [kg]} (1) \text{ ans}$ $31.6 \times 8.9 \times 10^{-2} (1) \text{ CO}_2 \text{ from 1 litre per week}$ $\times 52 = 146.24 (1) \text{ CO}_2 \text{ in a year from 1 litre per week}$ $\times 10 = 1462.4 (1) \text{ CO}_2 \text{ in a year from 10 litres per week}$ NOTES $\times 10 \text{ and } \times 52 \text{ seen anywhere } (1)$ $52 \times 10 \times 8.9 \times 10^{-2} = 46.28 \rightarrow 1 \text{ only}$		3		3	3	
	(b)			Reduce idling losses by having stop-start [system / buttons] (1) Reduce inertial losses by using low mass materials or regenerative braking (1) Reduce rolling resistance by using materials – or narrower tyres - which don't heat up as much as they are distorted or automatic tyre pressure warning system(1)	3			3		
	(c)			Calculation of KE: for 30 m/s = 522 000 J (1) KE for 10 m/s = 58 000 J (1) KE lost = $\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = 522\,000 - 58\,000 \text{ J}$ KE lost = 464 000 [J] ecf on a slip (1) Energy transferred = $0.6 \times 464\,000 \text{ (ecf on an energy value) } = 278\,400 \text{ [J]} (1)$ NB using 20 m/s with $\frac{1}{2}mv^2 \rightarrow 232\,000 \text{ J } [\rightarrow 0 \text{ marks}]$ $\rightarrow \text{Energy transfer } 139\,200 \text{ J } [\rightarrow 4\text{th mark}]$	1	1 1 1		4	4	
				Question 5 total	4	6	0	10	7	0